

Muhammad Ibrahim Umar

Lahore, Pakistan | u2024413@giki.edu.pk | Portfolio | LinkedIn: muhammad-ibrahim-umar | GitHub: Ibrahim1572

Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

2024 – 2028

Bachelor's Degree, Artificial Intelligence (Junior – 3rd Year)

Topi, KPK, Pakistan

Technical Skills

Languages: C++, Python, JavaScript/TypeScript

AI / ML: PyTorch, scikit-learn, NumPy, Pandas, NLTK, SVM, K-Means, PCA, Linear Regression, Generative AI (Gemini API)

Web & APIs: FastAPI, Docker, Next.js, React, TypeScript, Flask, MongoDB, JWT, OAuth 2.0, Tailwind CSS, Firebase

Data & Visualization: Plotly, Recharts, Altair, Seaborn, Jupyter Notebook, Streamlit

Frameworks & Tools: Qt Framework, SFML audio, PostgreSQL

Core Concepts: Data Structures and, Algorithms, OOP, Image Processing, Digital Logic Design, Software Documentation, SQL, Git, Git Hub

Experience

Software Engineering Intern — CyberNest (Lahore, Pakistan)

Jun 2026 – Aug 2026

- Built a production-grade full-stack content platform featuring hybrid authentication (credentials + Google/GitHub OAuth), RBAC, TOTP 2FA, and Edge-based silent JWT refresh middleware using Next.js 16, React 19, and MongoDB.
- Implemented an audit-logged content lifecycle engine with soft-delete/restore functionality and a time-series analytics dashboard.
- Independently rebuilt the entire system in Flask and React (Vite) to validate the architecture's portability across frameworks.

Projects

Philosophical Text Engine – Python, PyTorch, scikit-learn, FastAPI, Docker

- * Built a hybrid ML pipeline to classify philosophical text passages sourced from Project Gutenberg into branches (e.g., Ethics) and specific schools of thought using a multi-tier Hierarchical MLP in PyTorch.
- * Engineered a hybrid inference system combining a deterministic SVM baseline, probabilistic Neural Network predictions, and unsupervised K-Means clustering to deliver multi-perspective analysis of input passages.
- * Applied TF-IDF vectorization (10,000 features) with NLTK preprocessing; used PCA to project passages into 2D space for visual analysis of philosophical “distance” between texts.
- * Exposed the pipeline via a FastAPI `/predict` endpoint with Docker containerization; supports dual inference modes (Fast/Full-accuracy) with serialized model artifacts via Joblib.

Hospify – Enterprise Hospital Management System – React, Flask, PostgreSQL, Firebase

- * Engineered a relational, ACID-compliant Hospital Management System with 7 role-based workflows (Doctor, Nurse, Pharmacist, Billing, etc.), enforcing business logic directly in PostgreSQL via PL/pgSQL stored procedures and autonomous triggers.
- * Implemented real-time clinical automation including vitals-breach alert triggers, automated pharmacy inventory decrementing, and JSONB audit logging capturing full before/after state on every mutation.
- * Integrated Google Gemini 2.5 Flash for synthetic diagnostic report generation rendered into formatted PDF lab reports; built hybrid cloud backup and disaster recovery via Firebase Firestore.

Anomaly Detection via Statistical Intensity Analysis – Python, NumPy, Pillow, Streamlit

- * Developed an anomaly detection system using first-principles statistical analysis instead of Deep Learning, addressing the rare event problem in computer vision where limited defect datasets make Neural Networks difficult to train.
- * Implemented a detection engine based on Mean and Standard Deviation; pixels flagged as anomalies when deviation exceeds threshold k . Developed isolated R/G/B channel maps to isolate specific defect types.
- * Optimized to $O(N)$ pixel complexity for real-time performance on low-end hardware. Achieved zero-shot detection of structural cracks without pre-labeled training data and millisecond-range inference latency.
- * Built an interactive Streamlit dashboard with a real-time k -value slider to adjust the detection threshold dynamically.

SmartSaver – Grocery Substitution Engine – FastAPI, Gemini API, Docker — Micathon 2026

- * Independently built a full-stack web application at Micathon 2026 that uses AI-powered OCR to scan grocery receipts, identify imported products, and recommend local alternatives to reduce consumer spending.
- * Integrated Gemini 2.5 Flash Vision API on the FastAPI backend to extract item details from receipt images; cross-referenced results against a curated substitutions.json database for local brand matching.
- * Prototyped a React 19 frontend with Recharts-powered interactive savings dashboards; implemented keyword categorization across 6 food groups and a surplus calculation engine.
- * Containerized the full stack with Docker Compose; implemented a sample-receipt demo mode enabling zero-setup live demonstrations without physical receipts.

Music Playlist Management System – C++, SFML, Data Structures and Algorithms

- * Developed a high-performance music management and playback system without external databases, building a custom engine using advanced Data Structures and Algorithms to minimize memory overhead.
- * Implemented AVL Trees for $O(\log n)$ lookups, a custom Polynomial Rolling Hash Table, and Merge Sort to rank Most Liked songs.
- * Designed a Graph Theory/BFS recommendation engine based on user listening patterns.

Leadership

IET on Campus – GIKI

Sep 2024 – Present

Head (promoted from Volunteer)

Topi, KPK, Pakistan

Shaukat Khanum Memorial Cancer Hospital & Research Centre

Ramazan 2026

Volunteer – Ramazan Fundraising Drive

Certifications

Business Communication (HP LIFE) | **Building with the Claude API (Anthropic)** | **Introduction to AI (Google)**